

Ganesh Gupta

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EXPERIENCE

Oracle | *Software Engineer, AI Platform*

Jun 2025 – Present

- Built a **distributed executor** running across **Docker** and **AWS Lambda (multiple platform types)**; **S3** artifact memoization eliminated redundant recomputation.
- Designed **Redis checkpointing** so interrupted distributed workflows resume from the last committed step, with durable **state recovery**.
- Wrote systems code in **C++** and **Go** for **scheduling**, **resource allocation**, and **fault-tolerant placement**; ran clusters with **InfiniBand/RoCE** networking.
- Built **SSE** monitoring and incident detection, cutting redundant API calls **90%** and **debugging** issues before user impact.

QureAI | *ML Infrastructure Engineer*

Mar 2026 – Present

- Cut **p99 latency 420ms to 140ms** across **50K+ daily requests** with async serving on **GCP Cloud Run**.
- Built **CI/CD rollback** and **failure alerting**; on **on-call rotation**, triaged distributed incidents end-to-end.
- Ran distributed training via **Slurm** HPC scheduling, sustaining utilization through **node loss**; built **Prometheus/Grafana** monitoring.

Nomura Research Institute Financial Technologies | *Software Engineer*

Sep 2019 – Aug 2023

- Operated **25+ C++/SQL-backed microservices** for **1M+ users** on **AWS**, sustaining **99.9% uptime** over 4 years.
- Profiled compute/memory bottlenecks and optimized batching and request scheduling across **Python/C++** services, raising throughput **2.4x**.
- Built **health checks**, **retry policies**, and **automated failure recovery** for **C++** serving; drove **root cause analysis**, cutting escalations **60%**.
- Cut nightly SQL aggregation from **8 minutes to 90 seconds** with **indexed views** and **batch partitioning**.

PROJECTS

Distributed KV Store / VPC Control Plane Simulator

github.com/ganeshhgupta/distributed-kv-store

- **Raft-consensus KV store** with **linearizable reads** and **log compaction**; VPC simulator models subnet routing and security groups.

LangFetch

tinyurl.com/LangFetch

- **LangGraph** agent decomposes multi-table questions into dependency-ordered **SQL subqueries**; deployable as **MCP** server.

RootCause AI

github.com/ganeshhgupta/rootcause-ai

- **LangGraph** agents route incident alerts to specialists and synthesize **root cause reports** for faster resolution.

EDUCATION

University of Texas at Arlington | *M.S. Computer Science, GPA 4.0/4.0*

Aug 2023 – May 2025

MCKV Institute of Engineering | *B.Tech Computer Science*

Jul 2016 – Jul 2020

SKILLS

Languages: C++, Python, Go, Java, SQL, TypeScript

Systems: Distributed Systems, Raft Consensus, Networking, Databases, REST APIs, Data Structures & Algorithms

Cloud & Infrastructure: AWS (Lambda, S3, EC2, RDS, SQS), GCP, Docker, Redis, Slurm, CI/CD, Prometheus/Grafana

Tooling: Debugging, Root Cause Analysis, Monitoring, HTML/CSS

CERTIFICATIONS

AWS Certified Developer Associate | GCP Professional MLOps Engineer